

sunpura

P1 Meter Reader User Manual

SMeter-RS06



Plug in, Power Your Life!

Product Overview

The SMeter-RS06 is a smart meter reader specifically designed to extend the WiFi wireless network data communication capabilities of P1 electricity meters that lack network connectivity. It can connect to devices via P1/HAN for data communication. By connecting to a router, this meter reader can integrate the device into a cloud-based IoT platform, supporting remote data monitoring, remote control, remote upgrades, remote maintenance, edge-side energy interconnection, and local debugging. It also supports multi-unit wireless networking and parallel operation with backflow prevention, making it a crucial control hub for residential energy management and monitoring systems. It boasts a data response time of up to 500 ms per transaction.

Installation Preparation

(a) Compatibility check

1. Electricity meter requirements: Must be equipped with a P1 data port.
2. System requirements:
 - Network: 2.4GHz WiFi (5GHz band not supported)
 - For users with older meters: A separate 5V/1A Type-C power adapter is required.

(b) Parts list

Part Name	Quantity	Illustrate
P1 electricity meter	1 unit	Includes an integrated P1 interface plug
Quick Guide	1 copy	Includes APP download code
Type-C data cable	1 root	Power supply for old meters
WiFi rod antenna	1 piece	2.4G antenna

Safety Precautions

1. Confirm whether the data output of port P1 of the meter is activated. If it is not activated, data cannot be collected. (P1 Port Activation: Some smart meters in Belgium, Denmark, Sweden, and Switzerland have a P1 port that your energy provider needs to activate remotely. Without activation, your P1 meter reader will not receive any data. You can usually request this through your provider's portal or via email.)
2. Avoid touching the device with wet hands during operation. Do not modify the P1 interface or power circuit; otherwise, the warranty will automatically expire.
3. The area around the electricity meter should be kept well-ventilated and dry, and the equipment should be kept away from heat sources (such as radiators).

Installation Steps

Hardware connection

1. Basic installation: Insert the device's P1 plug into the P1 port of the smart meter (usually an RJ12 interface, labeled "P1" or "DATA").
2. Power supply configuration:
 - New meter (DSMR 5.0+): The P1 port can be directly powered, and the device's LED status light will remain on after insertion.
 - Older models (DSMR 4.0 and below): Connect the device to the power adapter using a Type-C data cable. The LED will light up when the power is on.
 - Regions with common P1 port support:

P1 ports are commonly found on smart meters in the following countries. While not every meter includes one, most modern models in these regions do.

Austria	Lithuania
Belgium	Luxembourg
Denmark	Netherlands
Finland	Sweden
Hungary	Switzerland



APP Connection, Configuration and Usage

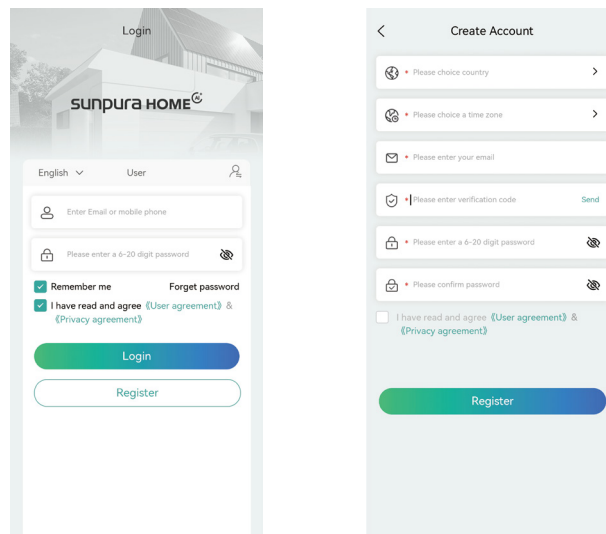
1. APP Download

- Scan the QR code below to access the Sunpura APP download navigation page; please select and download Sunpura APP according to your needs:



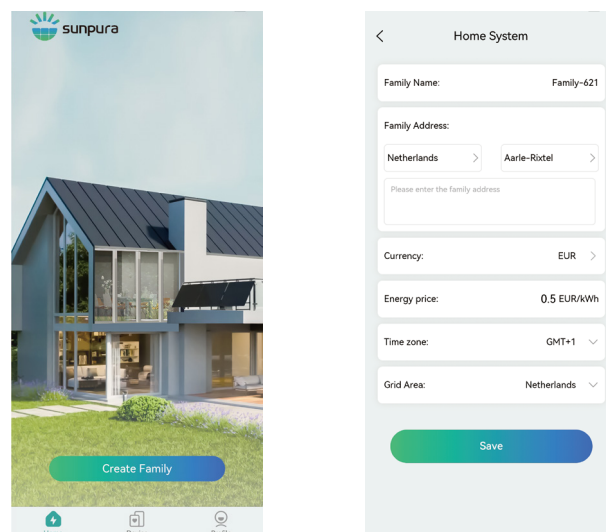
2. Registration process

- Run the Sunpura APP, go to the registration page, fill in the registration information as prompted, and click Register.



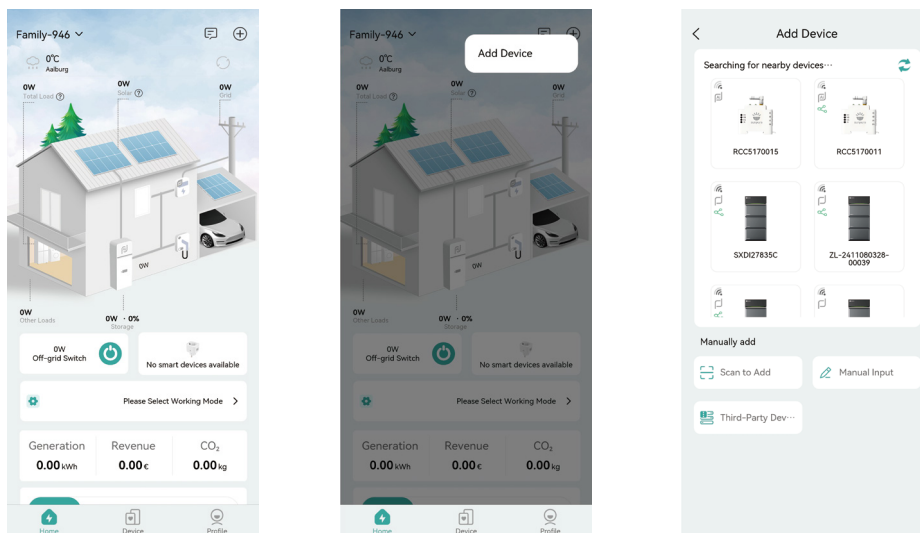
3. Add a power station

- Enter the app and click "Create Family" to enter the add power station page.
- Fill in the power station information as prompted.
- Click "Save" to add the power station successfully.



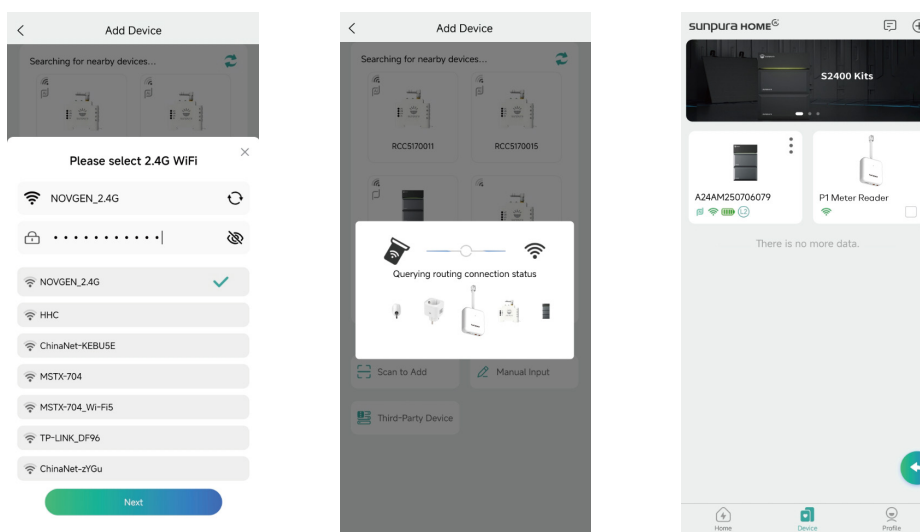
4. Add device

- Please ensure that location and Bluetooth permissions are enabled on your phone and the Sunpura app.
- Click the plus button in the upper right corner of the "Home" page, then click "Add Device" to enter the device adding page.
- Click the device icon or manually enter the device serial number to add the device.
- Once the Bluetooth connection is successful, you can proceed to step five.



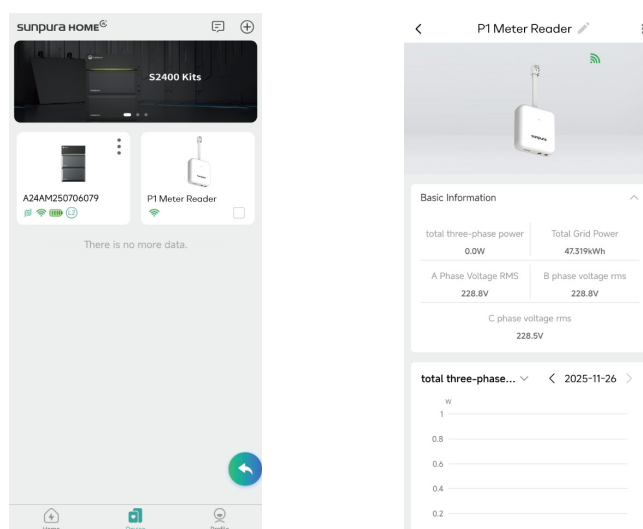
5. Network mode configuration

- Select or enter the WiFi name and password to connect to the network.
 - Click "Next" to begin network configuration. Please wait a few seconds for the configuration to complete successfully.
- Note: If network pairing fails, please check if your phone's location services and Bluetooth permissions are enabled. If they are correct, please follow the prompts again.

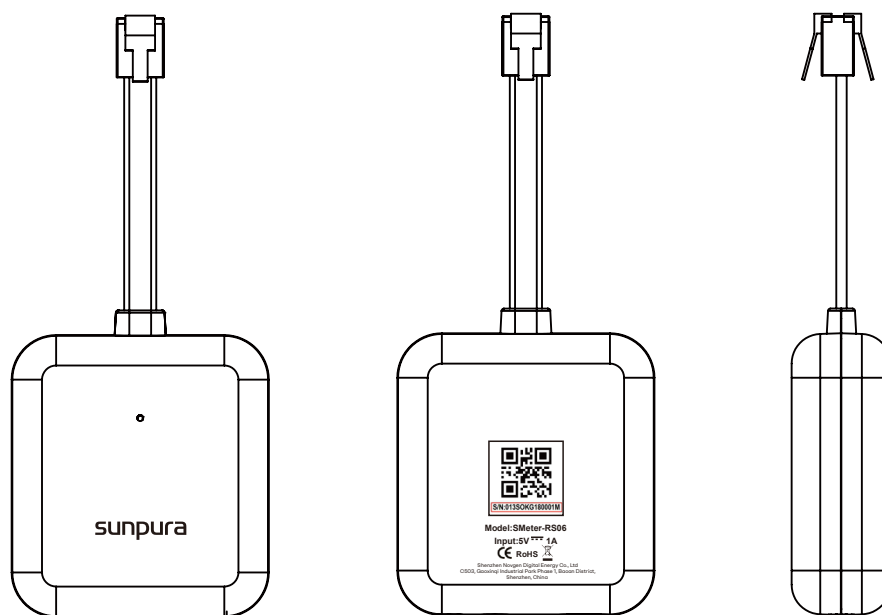


6. Check the status of the equipment

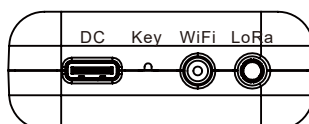
- Click the device icon to enter its details page. Scroll down to refresh the page. If the Wi-Fi icon in the upper-right corner of the image is green, it indicates "Online", meaning the device is connected and functioning normally. If it appears gray, indicating "Offline", the device has lost its network connection.



Product appearance, silkscreen printing, and dimensions



Front: Indicator light, Sunpura logo; Back: SN QR code, product parameters, manufacturer address



Bottom: From left to right, these are the DC power supply port, reset button, WiFi antenna mounting port, and LoRa antenna mounting port (reserved port, closed by default).

Technical Specifications

Technical parameters

Part Name	Project	Illustrate
General Parameters	Appearance (Length/Width/Height)	121 * 51.8 * 20mm
	Weight	45g
	User Manual Language	English
	Power supply	DC 5V
	Operating current	1A
	Power consumption	2.5W
	Operating temperature	-20°C~+65°C
	Storage temperature	-30°C~+85°C
	Installation method	RJ12 interface / RJ45 interface
	Certification	CE/RoHS
Interface	Warranty	2 years (hardware designed for a lifespan of 10 years or more)
	Key	1
	P1	1
	HAN	1
Show	Led	1
Equipment Management	Number of devices under management	1
	Communication methods	P1
	Network topology	WiFi
	Internet access methods	WiFi
	Energy Interconnection	Support
Wireless parameters (system & frequency band)	WiFi operating frequency	2400~2483.5 MHz
	WiFi wireless standard	802.11b/g/n
	WiFi speed	Up to 150 Mbps
	BLE	BLE 4.2 and above, 2.4 GHz
	WiFi & BLE communication distance	50m & 10m
Application Parameters	P1 baud rate	115200 bps (default)
	Supported servers	smartServer
	User Configuration	App
	Data upload cycle	3s

Indicator light description

	The electricity meter is not connected.	1. WiFi connection: Green light flashes quickly for 200 ms. 2. WiFi connection not established, server not connected: Green light flashes slowly for 1 second. 3. WiFi connection, server connection: The green light is always on.
	Electricity meter link	1. WiFi not connected: Blue light flashes quickly for 200 ms. 2. WiFi connection not established, server not connected: Blue light flashes slowly for 1 second. 3. WiFi connection, server connection: solid blue light.
	P1 fault	Red light on

Troubleshooting

Part Name	Solution
The status light is not on.	1. Check that the P1 port is properly inserted; 2. The old meter must be connected to a Type-C power supply; 3. Test with a different power adapter.
The app cannot recognize the device.	1. Confirm WiFi is 2.4 GHz; 2. Restart the router; 3. Reset the device (press and hold the reset button for 30 seconds).
Data update interrupted	1. Check if the P1 port of the electricity meter has been shut down by the power grid company; 2. Update the app to the latest version.
Electricity price calculation error	1. Verify the tax rate settings in [Intelligent Mode]; 2. Update the regional electricity price (the app automatically pushes updates monthly).

Precautions

1. The device requires regular firmware upgrades (the app will automatically prompt you; power must not be interrupted during upgrades).
2. If you replace the electricity meter, you need to add the device again in the app and synchronize the power grid company information.
3. Data storage complies with the EU GDPR.

Contact Us

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